BAYERO UNIVERSITY KANO

FACULTY OF ENGINEERING,

DEPARTMENT OF MECHATRONICS ENGINEERING.

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PROJECT TITLE:

AN AUTO EVACUATION SYSTEM FOR BASEMENT WASTE

BY

GROUP 3

MEMBERS:

NAME	*REG NUMBER*
FUAD-NASIRUDEEN ABDUSSALAM	ENG/20/MCT/00301
OLALEKAN ABDULFATAI LANRE	ENG/20/MCT/00306
SALIHU ABDULKADIR	ENG/20/MCT/00308
YAHYA ABDULMUIZ ISHOLA	ENG/20/MCT/00310
YAKUBU ABDURRAHMAN BABA	ENG/20/MCT/00311

Supervised by: Engr. A. A. Lawal

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May Almighty Allah reward everyone abundantly.

CERTIFICATION

This is to certify that the group project titled "An Auto Evacuation System for Basement Waste" was carried out by the following students of Mechatronics Engineering Department, Faculty of Engineering, Bayero University Kano, in partial fulfilment of the requirements for MCS4209 (Group Project) in Department Mechatronics Engineering.

1. Abdussalam Fuad-Nasirudeen Ayodeji – ENG/20/MCT/00301
2. Olalekan Abdulfatai Lanre – ENG/20/MCT/00306
3. Salihu Abdulkadir – ENG/20/MCT/00308
4. Yahya Abdulmuiz Ishola – ENG/20/MCT/00310
5. Yakubu Abdurrahman Baba – ENG/20/MCT/00311

_____

Engr. A. A. Lawal
Supervisor

Date: _____

ABSTRACT

This project presents the design and implementation of an Auto Evacuation System for basement waste. The system addresses the challenge of managing waste in underground septic tanks, particularly when the soak away is positioned at a higher level or where the water table is close to the base of the septic tank. To prevent overflow and chemical degradation of the tank walls, an automatic evacuation mechanism was developed.

The system comprises two pumps, an Arduino Uno microcontroller, ultrasonic sensor, Relays, an LCD display, connectors, and plastic containers representing the septic tank and soak away. The Arduino continuously monitors the liquid level via the ultrasonic sensor and activates the first pump when the waste reaches a predefined threshold. In case of pump failure, a backup pump is automatically triggered, and an error message is displayed on the LCD. Additionally, manual control via a push button is provided for testing and maintenance purposes.

The prototype demonstrates a reliable and efficient approach to automating waste evacuation from basement septic systems, minimizing health risks, and enhancing environmental safety.

INTRODUCTION

1.1 Background

In modern building infrastructure, especially those incorporating underground facilities such as basements, efficient waste management poses a significant engineering challenge. Unlike ground-level systems, basement sewage systems operate below the natural drainage gradient, making gravitational flow to the soak away or main sewage line impossible. In such conditions, waste stored in a septic tank cannot be discharged automatically due to its lower elevation relative to the soak away, resulting in frequent manual evacuation and potential environmental hazards.

Furthermore, when the base of a septic tank is positioned close to the water table, prolonged retention of waste can lead to chemical reactions capable of weakening or cracking the walls of the tank. This leakage may contaminate the surrounding groundwater and cause structural damage to the building foundation. Therefore, a reliable automated system that can monitor the waste level and safely evacuate it to the soak away becomes an essential solution for sustainable waste management in basement environments.

The **Auto Evacuation System for Basement Waste** was designed to address this problem by integrating sensing, control, and pumping mechanisms into a fully automated unit. The system uses an Arduino microcontroller to continuously monitor waste levels through an ultrasonic sensor and to activate a pump (or a backup pump) whenever the liquid level exceeds a preset threshold. The inclusion of an LCD display provides real-time status updates, while manual override buttons enable maintenance checks and emergency operation.

This approach not only simplifies basement waste evacuation but also ensures hygiene, prevents overflow incidents, and extends the lifespan of the septic structure by minimizing chemical deterioration due to prolonged waste retention.

1.2 Problem Statement

In many buildings with basements, the soak-away system lies at an elevation higher than the basement floor. As a result:

- Gravity flow from basement plumbing to the soak-away is impossible.
- Manual emptying of the basement septic tank is laborious, time-consuming, and may be delayed or forgotten, risking overflows.
- Without automated monitoring, the system may fail or overflow if the pump does not operate exactly when needed.
- Prolonged retention of waste can lead to chemical reactions capable of weakening or cracking the walls of the tank

These challenges motivate the need for a reliable solution that can

1. monitor waste level
2. trigger pumping when a threshold is reached, detect pump failure and switch to a backup pump
3. operate with minimal human intervention.

By developing a model system, one can demonstrate how automation can reduce maintenance, prevent overflows, and increase system robustness.

1.3 Project Objectives

The main objectives of this project are as follows:

1. **Design and build** a small-scale model of an automatic septic tank evacuation system tailored for a basement scenario, using readily available materials (e.g., plastic containers) to represent a septic tank and soak-away.
2. **Implement level sensing** via an ultrasonic sensor to measure effluent height in the "septic tank" and trigger the evacuation process when a threshold is exceeded.
3. **Incorporate redundancy and reliability**, by using two pumps—one primary and one backup—such that, if the primary pump fails, the system automatically switches to the backup pump.
4. **Develop control logic and software** (e.g., on an Arduino microcontroller) that manages level sensing, pump activation, failure detection, and transition to backup mode.
5. **Evaluate functionality and performance**, demonstrating that the system reliably evacuates effluent when the level threshold is reached, and handles pump failure gracefully.
6. **Highlight the importance and potentials** of automating septic/effluent systems in constrained building environments (e.g. basements) to reduce manual work, minimize failures, and improve safety and sanitation.

LITERATURE REVIEW

2.1 Introduction

Waste management in building infrastructure has evolved over the years from entirely manual systems to semi-automated and now fully automated systems. The major focus of these systems is to ensure efficient evacuation, minimize environmental pollution, and reduce human involvement in waste handling. In basements and underground facilities, traditional gravity-based systems are ineffective due to the lower elevation of the septic tank relative to the soak away, hence the need for an intelligent pumping control mechanism.

Several researchers and engineers have proposed systems utilizing microcontrollers, sensors, and motorized pumps to automate fluid management. Ultrasonic sensors are widely used for non-contact level measurement due to their accuracy, affordability, and ability to operate effectively in wet or corrosive environments. Similarly, microcontrollers like the Arduino platform provide a flexible and programmable interface to control pumps, display systems, and alarms based on sensor feedback.

However, most existing designs focus primarily on water level control in overhead tanks or irrigation systems, rather than basement septic tank evacuation, where the fluid involved is waste material that may contain solids, gases, and corrosive chemicals. This project builds upon previous research by integrating redundancy (dual pumps), fault detection, and user notification into a single low-cost automated system.

Thus, this literature review focuses on exploring existing techniques for liquid level monitoring, automatic pump control, microcontroller-based waste systems, and safety integration — with the aim of identifying the most suitable methods for designing a reliable and efficient Auto Evacuation System for Basement Waste.

2.2 Discuss the literature of your Project

The need for efficient waste evacuation systems has long been recognized in civil and environmental engineering. Over the years, researchers and engineers have developed several automated systems aimed at monitoring and controlling fluid levels in tanks, wells, and drainage systems. These works have provided valuable insights and form the theoretical and technical foundation upon which this project was developed.

Automatic Water Level Control Systems

Early research in liquid-level automation focused mainly on domestic water supply systems. Several studies introduced microcontroller-based water level controllers that used float switches, probes, or ultrasonic sensors to regulate the operation of pumps automatically.

For instance, Joseph et al. (2018) developed an Arduino-based automatic water level controller that monitored tank water levels using an ultrasonic sensor and controlled a motor pump accordingly. Their system displayed the level on an LCD and turned the pump on or off based on preset thresholds. Similarly, Patel and Singh (2019) implemented a low-cost water tank monitoring system using an HC-SR04 ultrasonic sensor and a relay-driven pump.

These systems proved efficient for water tanks positioned above ground level but were not optimized for waste systems, where the nature of the liquid (semi-solid and corrosive) demands stronger components and continuous reliability.

Automated Sewage and Drainage Systems

Recent studies have extended the concept of automation into waste evacuation and sewage management. Kumar and Mehta (2020) proposed a microcontroller-based sewage level monitoring system that triggered an alarm when the sewage level reached a critical point. Their work highlighted the importance of early overflow detection to prevent environmental contamination. However, it relied solely on audible alerts and lacked an automatic pumping mechanism. Rahman et al. (2021) presented a smart drainage cleaning system using Arduino and ultrasonic sensors. The system automatically opened and closed drainage gates depending on water levels, demonstrating how automation could minimize flooding in low-lying areas. Though similar in principle, it was not tailored for septic tank evacuation or pump control redundancy.

Integration of Backup Pumping and Fault Detection

In industrial and building automation, the concept of redundant control is critical for ensuring system reliability. Studies by Singh et al. (2022) introduced a dual-pump water transfer system that used feedback sensors to detect pump malfunction and switch automatically to a backup unit. This approach greatly inspired the design of the current project, which also employs two pumps — one primary and one standby — to maintain continuous evacuation even during mechanical or electrical failure of the main pump.

Gaps in Existing Systems

Although prior works on liquid level automation have achieved significant progress, most systems are designed for clean water environments, not wastewater or basement septic applications. Many do not address the challenges of:

- Operating under low-level (underground) conditions where gravity cannot assist flow.
- Handling semi-solid waste with possible pump blockage.
- Detecting and responding to pump failure automatically.
- Providing user alerts for system malfunction or overflow.

Contribution of the Present Work

The Auto Evacuation System for Basement Waste improves upon existing systems by combining automatic sensing, dual-pump redundancy, time-based flushing, error display, and manual override into a single prototype. It demonstrates a practical model for smart waste evacuation systems suitable for basements or low-lying areas. The system uses the Arduino Uno as the control unit, an ultrasonic sensor for level measurement, and two pumps for redundancy — ensuring safety, efficiency, and reliability.

2.3 Microcontroller and Sensor Technology

The Arduino Uno microcontroller has been a central component in many small-scale automation projects due to its simplicity, cost-effectiveness, and adaptability. It supports integration with various sensors, relays, and output modules.

Ultrasonic sensors, specifically the HC-SR04, have become the preferred choice for non-contact level measurement because they are accurate, immune to contamination, and suitable for liquids that are not optically clear.

The Liquid Crystal Display (LCD) is commonly used to provide user feedback, displaying real-time system status, level readings, and error messages. These features enhance usability and monitoring efficiency.

2.4 Relevant Past Projects

This chapter reviews prior work and key technical concepts that inform the design of an automatic septic tank evacuation system for a basement. It covers

1. Pumped septic / STEP (Septic Tank Effluent Pumping) systems and design guidance
2. Common level-sensing technologies and their trade-offs
3. Microcontroller-based automation (Arduino prototypes) used for tank monitoring and pump control
4. Redundancy and dual pump control strategies used in practice
5. Decay, retention time, and time-based flushing.

Septic Tank Effluent Pumping (STEP) systems

When gravity discharge is not possible for example, when a basement's outlet is below the soak-away elevation, septic tank effluent pumping (STEP) or pumped effluent systems are the standard engineering solution. Sewage is collected in the tank and an effluent pump periodically discharges liquid to the soak-away under pressure. Design guidance for STEP systems emphasizes correct sizing, protected discharge piping, reliable pump selection, and access for maintenance. These guidelines also recommend record-keeping and acceptance testing to ensure long-term performance.

Takeaway for our project: a small-scale model that collects effluent in a tank and uses a pump to lift and discharge it directly reflects real STEP system principles (intermittent pumping, attention to discharge piping and valves, and need for reliability).

Level sensing options for septic tanks — trade-offs and suitability

Several common technologies measure liquid level in tanks; choice depends on cost, environment (sediment, gases), accuracy needs, and mounting:

- Float switches (mechanical or reed): cheap and simple, but prone to fouling and mechanical failure in septic environments.
- Ultrasonic sensors: non-contact measurement that sends acoustic pulses and measures echo time to compute distance to liquid surface. Advantages include no direct contact with wastewater (less fouling), reasonable accuracy, and easy interfacing to microcontrollers.
- Radar (microwave) sensors: similar non-contact approach; better in vapor/foam/turbulence but typically costlier than ultrasonic for small projects.

Takeaway for our project: For a low-cost prototype the ultrasonic sensor offers a good balance of accessibility and functionality. Its non-contact nature suits a septic model where solids and gases would foul contact sensors limitations (foam/turbulence) can be mitigated in a lab model by sensor positioning or using a stilling pipe. Relevant case studies and hobby/academic prototypes report success using ultrasonic sensors with Arduino for septic or wastewater level monitoring.

Microcontroller-based automation (Arduino prototypes and control strategies)

There is extensive hobbyist and academic work showing that Arduino (or similar microcontrollers) can reliably read level sensors and control pump relays. Typical implementations:

- Read ultrasonic (or pressure/float) sensor input → apply threshold logic → drive a relay or motor controller to switch the pump on for a pumping cycle or until level drops below a lower threshold.
- Add timing, debouncing, and hysteresis to avoid pump chatter (short on/off cycles).
- Log or alarm status (LED, buzzer, or GSM/SMS) for high-water or pump failure conditions.

Published prototype papers and project posts show ultrasonic sensor + Arduino + relay modules are a common pattern for automatic pumping or tank monitoring, including adaptations for septic environments (e.g., adding MQ gas sensors, remote alerts).

Takeaway for our project: Arduino provides straightforward I/O and programming for implementing level thresholds, pump activation, and simple failure detection and pump switching—making it appropriate for a demonstrator model and for students to explore control logic and fault handling.

Redundancy: dual pump arrangements and control

In real pump stations and STEP installations, redundancy is a key design consideration where failure has serious consequences (overflows, health hazards). Dual pump (primary + backup) and control are standard industry solutions: the control logic typically alternates duty between pumps, detects pump faults (overcurrent, lack of expected level change), and activates backup pumps on failure, often with alarms to notify operators. Engineering sources emphasize that redundancy reduces downtime and prevents overflows, and good control panels include alarms, fuses, check valves, and easy pump removal for maintenance.

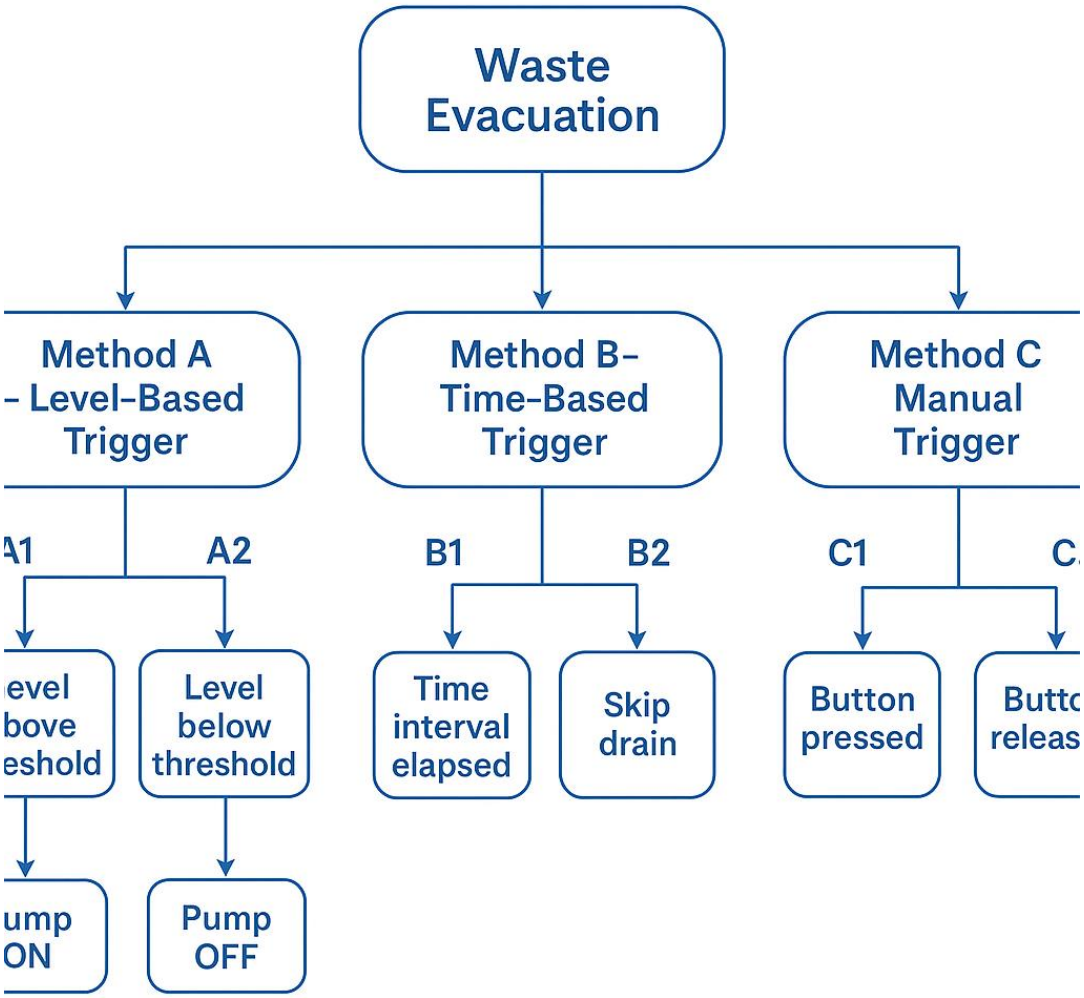
Takeaway for our project: Including a backup pump in the model mirrors professional practice and allows demonstration of automated failure detection and switching logic (e.g., monitor pump current or timeout on expected level drop → activate backup pump + alarm).

Waste Decomposition and the Need for Time-Based Flushing

In addition to monitoring liquid levels, a septic system must account for biological waste decay when the system is left unused for long periods. Inside a septic tank, anaerobic bacteria decompose organic matter into simpler compounds such as methane, carbon dioxide, and water. Studies and engineering guidelines report that decomposition begins within 24–72 hours after deposition, with about 40–50 % of solids biologically digested over time, while the remainder accumulates as sludge or scum layers. Although this process is gradual, long retention without discharge can cause odor buildup, gas corrosion, or blockage—especially when the tank is nearly full.

Takeaway for our project: Our proposed model includes a time-based flushing feature if no evacuation occurs after a preset interval in order to prevent accumulation of stagnant waste which can cause hydrogen-sulfide corrosion, sludge hardening, and microbial imbalance.

2.5 Feedback Control



A. Feedback Control Principle

The Auto Evacuation System for Basement Waste is fundamentally based on the feedback control principle, which enables automatic regulation of the waste level in the septic tank. In a feedback system, the output (in this case, the waste level) is continuously monitored and compared with a desired set point. Any deviation from the expected condition is detected by the sensor and used to generate corrective action.

In this project, the ultrasonic sensor measures the distance between the liquid surface and the sensor head. This information is fed into the Arduino microcontroller, which processes the signal and determines whether the level has exceeded or fallen below the defined threshold. Depending on the feedback, the microcontroller switches the pump(s) ON or OFF through the RELAY modules.

This continuous sensing and correction process ensures that the waste never exceeds the safe limit or remains stagnant for too long. The system's decision-making logic therefore represents a closed-loop control process, as illustrated in Figure above.

B. System Automation and Control Logic

Automation in this project is achieved through embedded programming in the Arduino Uno microcontroller. The code defines threshold levels such as the start-drain level (70%) and stop-drain level (20%) of the tank's height. When the measured level crosses these limits, the control logic automatically initiates or terminates pumping actions.

A timed operation is also integrated, allowing the system to drain the tank at regular intervals (for example, every 3 minutes) even if the level sensor does not trigger. This redundancy prevents waste stagnation in low-usage conditions.

Additionally, the system incorporates fault detection by monitoring the level response after the pump is activated. If the level does not decrease as expected, the controller assumes that the primary pump has failed and automatically activates the secondary pump. The LCD screen then displays an error message to alert the user of the malfunction.

This multi-level control approach improves system reliability and ensures that evacuation continues uninterrupted, even in the event of a partial system failure.

C. Comparative Analysis with Related Systems

Compared to earlier designs of automatic water level controllers or sump pump systems, this project provides a higher degree of autonomy and fault tolerance. While most previous designs focus solely on activating a single pump based on level detection, this system integrates:

- Dual-pump redundancy for reliability.
- Automatic and time-based control.
- Real-time feedback display.
- Manual override functionality.

These features make it more adaptable to real-world scenarios where environmental, mechanical, and electrical factors can affect system performance.

D. Summary

This section has discussed the control principles and operational logic that govern the Auto Evacuation System for Basement Waste. By integrating feedback sensing, automated decision-making, and redundant actuation, the system achieves robust, safe, and continuous waste evacuation for basement septic tanks. The next chapter will describe the methodology used in implementing the hardware and software aspects of the project.

METHODOLOGY

3.1 Introduction

This chapter explains the methods and design approach used in developing the Automatic Septic Tank Evacuation System for a Basement. The project follows a systematic engineering design process — from problem definition, through component selection, circuit design, coding, and system testing. The aim is to construct a small-scale model that automatically evacuates waste from a basement-level septic tank to a soak-away whenever the waste level exceeds a set threshold, and to include additional reliability features such as a backup pump and a time-based flush module.

The entire methodology is divided into the following sections:

- Design methodology
- Project tools and cost
- Hardware design
- Software design
- Time-based flush module
- Personal contribution

3.2 Design Methodology

The design methodology of the Auto Evacuation System for Basement Waste involves both hardware and software integration to achieve a fully automated and reliable waste transfer system. The approach follows a modular design, where each part of the system performs a specific function and communicates with other modules through a central control unit — the Arduino Uno microcontroller.

The system was modeled and implemented in a scaled-down prototype form to simulate real-world operation. Two transparent plastic containers were used to represent the septic tank and soak away, respectively. The septic tank container was fitted with an ultrasonic sensor mounted on the top to measure the liquid level. Two DC pumps were placed near the base of the septic tank container, each connected to a separate Relays and responsible for transferring waste to the soak away container through flexible plastic tubes.

The first pump acts as the primary evacuation unit, while the second serves as a backup that automatically activates when the first fails. A 16×2 LCD display connected to the Arduino provides real-time feedback such as waste level percentage, pump operation status, and error messages. A manual push button allows the user to trigger the pumps for testing or emergency evacuation, ensuring full control flexibility.

The Arduino Uno serves as the main control unit that processes sensor data and executes logic decisions. The ultrasonic sensor (HC-SR04) measures the distance from the sensor to the waste surface. This value is converted into a Percentage level of tank fullness using programmed calculations. When the level exceeds the defined upper limit (70%), the controller automatically activates Pump 1. Once the level drops below the lower limit (20%), the pump is deactivated to prevent dry running. “The activation threshold of 70% and minimum stop level of 20% were chosen respectively to reduce

frequent pump cycles and prevent pump overworking. This adjustment improves pump longevity and ensures a more stable evacuation cycle while still preventing overflow."

A time-based trigger was also included, allowing the system to initiate evacuation periodically (every few minutes) even if the liquid level remains constant — preventing stagnation and chemical buildup inside the tank. If Pump 1 fails to reduce the level after activation, the Arduino identifies the fault, deactivates Pump 1, and starts Pump 2.

The design flow can be summarized as follows:

1. The ultrasonic sensor continuously monitors the liquid level.

2. The Arduino reads and converts the sensor data into percentage values.

3. The control logic determines if the level exceeds or falls below preset thresholds.

4. The appropriate pump(s) are activated or deactivated based on the level.

5. The LCD display shows real-time system information and error messages.

6. Manual input can override the automatic mode when required.

7. Time-Based Trigger

The entire system was powered by a regulated DC supply suitable for Arduino and pump operation. During testing, the system demonstrated reliable automatic operation, proper switching between pumps, and accurate level detection within the limits of the prototype setup.

3.3 Project Tools and cost

S/N	ITEM	QTY	COST (₦)	TOTAL (₦)
1	Arduino Uno Microcontroller	1	25000	25000
2	Ultrasonic Sensor (HC-SR04)	1	2500	2500
3	16x2 LCD with I2C module	1	5000	5000
4	Push Button	1	300	300
5	Relay Module	2	2500	5000
6	DC pumps	2	2000	4000
7	Breadboard	1	1000	1000
8	Water Hose	1	1000	1000
9	Plastic Containers	3	1500	4500
10	Miscellaneous	-	2000	2000
			Total	50300

3.4 Hardware Design

The hardware design of the Auto Evacuation System for Basement Waste integrates several electronic components and electromechanical devices into a single automated unit. The hardware section consists of the Arduino Uno microcontroller, ultrasonic sensor (HC-SR04), two DC pumps, two relay modules, LCD display, manual push button, connecting wires and power supply.

The system is implemented as a scaled-down prototype using two transparent plastic containers to represent the septic tank and soak away. The ultrasonic sensor is mounted at the top of the septic tank container and connected to the Arduino's digital pins (trigger on pin 6, echo on pin 5). This sensor measures the distance from the sensor head to the liquid surface and converts it into a proportional level value. The readings are used by the microcontroller to determine whether the liquid has reached the maximum or minimum threshold.

At the base of the septic tank container, two small DC pumps are installed. Each pump is responsible for transferring waste to the soak away through individual pipes. The first pump acts as the primary evacuation pump, while the second acts as a backup unit that is automatically triggered if the first fails to operate properly. The pumps are powered through two separate relay modules, which provide the necessary current and voltage required to control each motor safely.

The Arduino Uno serves as the brain of the system. It continuously receives input data from the ultrasonic sensor and, based on programmed logic, sends output control signals to the motor driver circuits through digital pins 3 and 4. When activated, the motor drivers energize the respective pumps, initiating the transfer of liquid from the septic tank to the soak away.

A 16×2 Liquid Crystal Display (LCD) module is connected to the Arduino through an I2C Module. It displays the current level of waste in percentage, the operational status of both pumps (ON/OFF), and any error or warning messages such as "Pump1 Failure" or "Level High". This visual feedback allows users to monitor the system's condition in real time.

For testing and maintenance purposes, a manual Push button is connected to the Arduino as an override switch. When pressed, it manually triggers the pumps irrespective of the sensor readings. This feature ensures that the user can test or evacuate waste manually when desired.

The entire circuit is powered by a regulated 12V DC supply, which is adequate for the Arduino board and DC pumps. Ground connections are common to all modules to ensure stable operation.

During the design and assembly stage, careful attention was given to component placement and electrical insulation, particularly around the pumps and liquid containers, to prevent short circuits or corrosion. Jumper wires and breadboard connections were used for prototyping, and the final circuit was tested under different simulated conditions to verify performance.

The overall interconnection of the system is illustrated in Figure 3.1, showing how each module communicates with the Arduino Uno controller.

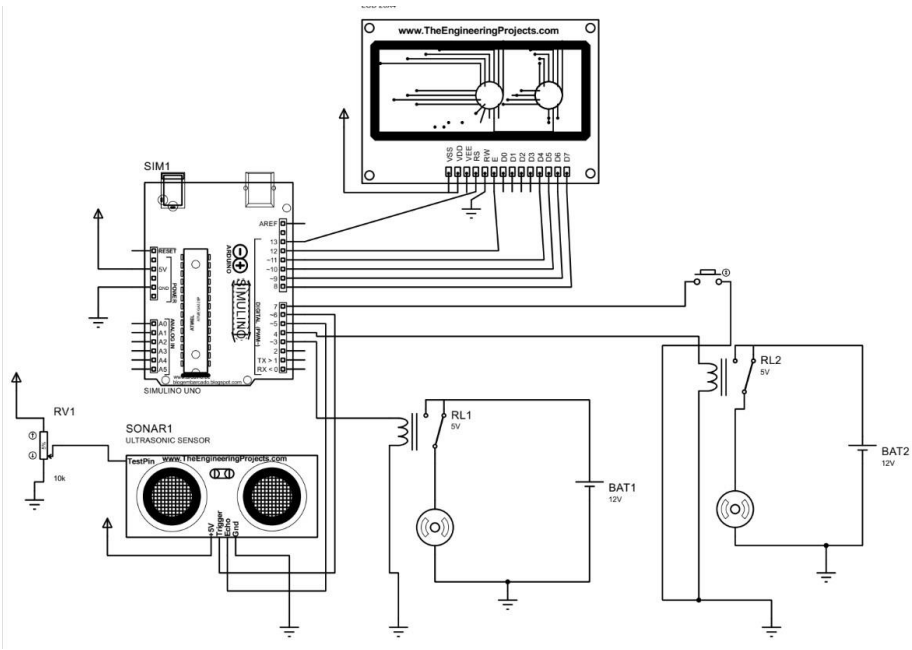


Fig 3.1: Schematic Diagram Of Electrical Connection of System

3.5 Software Design

Proteus Design Suite (Version 8.17)

This electronic design automation (EDA) software was used extensively in the initial design phase. Its mixed-mode simulation capability allowed for the creation of a virtual prototype of the entire electronic circuit. The Arduino Uno model, HCSR04 sensor model, and virtual components were placed and connected within the Proteus ISIS schematic capture module. This virtual environment was crucial for:

- **Logic Verification:** Testing the program's flow and the interaction between the Arduino, the sensor input, and the digital output to the relay without any physical hardware.
- **Circuit Debugging:** Validating the correctness of the circuit design, including pin connections and the use of the relay module, before using a breadboard.

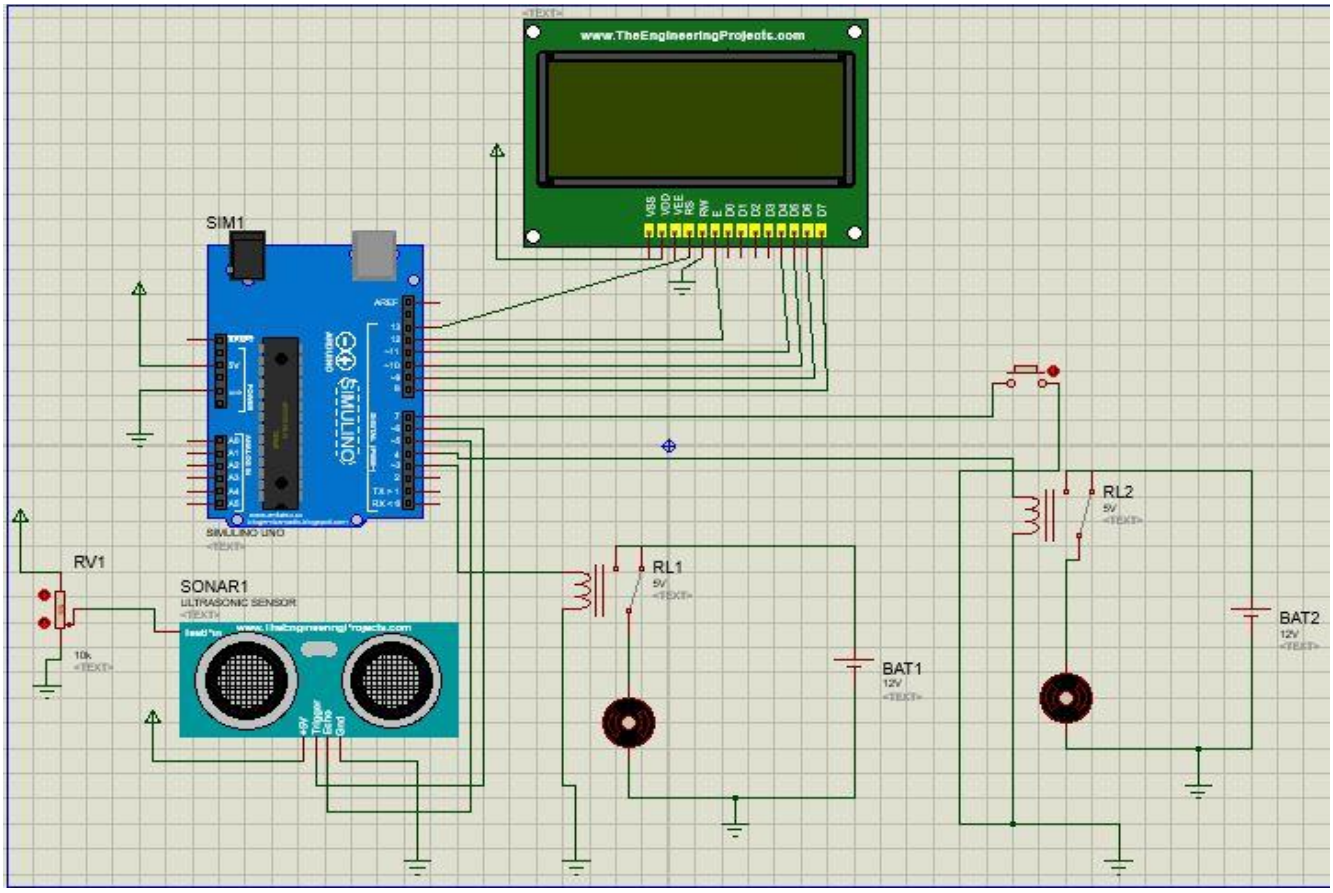


Fig 3.2: Schematic Diagram of Electrical Connection of System

Simulation Setup in Proteus Professional Suite

- Placed the Arduino Uno model from Proteus library to the schematic capture window.
- Placed the ultrasonic sensor model with defined TRIG and ECHO terminals in proteus library to schematic capture window.
- Placed a Pushbutton, LCD, Battery Power Source and a SPDT relay from proteus library to schematic capture window (the connection of the relay, transistor and diode is used as the motor driver).
- Placed the DC Pumps which was represented in proteus using DC Motor.
- Connected all components as described in the Proteus Schematic diagram above.
- Load the compiled .hex file of the Arduino sketch into the Arduino model in the Proteus Design Suite.
- Simulated the design to confirm the functionality of the design.

Arduino IDE (Version 2.3.2)

The Arduino Integrated Development Environment was used to write, compile, and upload the firmware to the Arduino Uno microcontroller in the Proteus Software and the physical Arduino Uno microcontroller board.

Attached below is the code written for the system in the Arduino IDE

Septic.ino

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>  //Include LCD Library

// LCD wiring
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Ultrasonic pins
#define TRIG 6
#define ECHO 5

// Pump relays
#define PUMP1 3
#define PUMP2 4

// Button Pin
#define button 7

// Tank parameters
const int tankHeight = 17; // cm
const int startDrainLevel = 70; // % full
const int stopDrainLevel = 20; // % full

bool pump1On = false;
bool pump2On = false;

// Automatic Drain Time
unsigned long lastDrainTime = 0;
const unsigned long drainInterval = 3L * 60L * 1000L; // 3 mins

```

Septic.ino

```
void setup() {
  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);
  pinMode(PUMP1, OUTPUT);
  pinMode(PUMP2, OUTPUT);
  pinMode(button, INPUT_PULLUP);

  digitalWrite(PUMP1, LOW);
  digitalWrite(PUMP2, LOW);

  lcd.init();
  lcd.backlight();
  lcd.print("Extraction Sys");
  delay(2000);  // Welcome screen
  lcd.clear();
}

float getLevelPercent() {
  // Measure distance
  digitalWrite(TRIG, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG, LOW);
  long duration = pulseIn(ECHO, HIGH);
  float distance = duration * 0.034 / 2; // cm

  // Convert to % full
  float level = tankHeight - distance;
  return constrain((level / tankHeight) * 100, 0, 100);
}

```

Fig 3.3: Screenshot of Code written on Arduino IDE

Septic.ino

```
void loop() {
  float levelPercent = getLevelPercent();
  unsigned long currentTime = millis();


  //Display status
  lcd.setCursor(0, 0);
  lcd.print("Level: ");
  lcd.print(levelPercent);
  lcd.print("%  ");
  lcd.setCursor(0, 1);
  if (pump1On)
    lcd.print("Pump1: ON");
  else if (pump2On)
    lcd.print("P1:Failed P2:ON");
  else
    lcd.print("Pumps: OFF ");

  delay(1000);


  // Time-based trigger OR Level-based trigger
  bool timeToDrain = (currentTime - lastDrainTime >= drainInterval);
  bool levelHigh = (levelPercent >= startDrainLevel);

  bool manualdrain = !
digitalRead(button);

```

Septic.ino

```
  if (levelHigh || manualdrain)  {
    if (!pump1On) {
      digitalWrite(PUMP1, HIGH);
      pump1On = true;
      delay(5000); // give Pump 1 a chance
      if (getLevelPercent() >= startDrainLevel) {
        // Pump 1 failed in a level-trigger situation
        digitalWrite(PUMP1, LOW);
        pump1On = false;
        digitalWrite(PUMP2, HIGH);
        pump2On = true;
      }
    }
    lastDrainTime = millis(); // reset timer after starting a drain

  } else if (timeToDrain) {
  // Timer-based drain logic
  if (levelPercent > stopDrainLevel) {
    if (!pump1On) {
      digitalWrite(PUMP1, HIGH);
      pump1On = true;
      delay(5000);
      if (getLevelPercent() >= startDrainLevel && !timeToDrain) {
        // Pump 1 failed in a level-trigger situation
        digitalWrite(PUMP1, LOW);
        pump1On = false;
        digitalWrite(PUMP2, HIGH);
        pump2On = true;
      }
    }
    lastDrainTime = millis();
  }
```

Septic.ino

```
        digitalWrite(PUMP2, HIGH);
        pump2On = true;
      }
    }
    lastDrainTime = millis(); // reset timer after starting a drain

  } else if (timeToDrain) {
  // Timer-based drain logic
  if (levelPercent > stopDrainLevel) {
    if (!pump1On) {
      digitalWrite(PUMP1, HIGH);
      pump1On = true;
      delay(5000);
      if (getLevelPercent() >= startDrainLevel && !timeToDrain) {
        // Pump 1 failed in a level-trigger situation
        digitalWrite(PUMP1, LOW);
        pump1On = false;
        digitalWrite(PUMP2, HIGH);
        pump2On = true;
      }
    }
    lastDrainTime = millis();
  }

}
  else if (levelPercent <= stopDrainLevel) {
  digitalWrite(PUMP1, LOW);
  digitalWrite(PUMP2, LOW);
  pump1On = false;
  pump2On = false;
}
  delay(1000);
}
```

Program Logic

The control software executes the following algorithm:

1. Read tank level from ultrasonic sensor.
2. If level ≥ high threshold → activate primary pump.
3. Monitor level drop; if none detected within timeout → activate backup pump.
4. Stop pump when level ≤ low threshold.
5. Track elapsed time since last pumping event.
6. If elapsed time ≥ Set maintenance time (e.g., 14 days or for the purpose of the presentation 3 minutes) → perform time-based maintenance flush.
7. Allow for Manual Flushing when Pushbutton is pushed.

3.6 Time-based Flush Module

Introduction

One limitation of conventional level-triggered pumping systems is that they remain inactive when the effluent level is below the sensor threshold. If the homeowners travel and the septic tank remains partly filled for weeks, the waste can stagnate and begin anaerobic decay, producing corrosive gases and odor. To mitigate this, a time-based flush control was incorporated into the Arduino program.

Design Concept

The microcontroller tracks the elapsed time since the last pumping event(T). If this period exceeds a defined maximum, the system performs a short maintenance-flush cycle, even if the level sensor has not triggered. This approach simulates periodic operation recommended in engineered STEP systems

Operation and Testing

During testing, the model was filled to half capacity and left idle to simulate homeowner absence. the Arduino automatically activated the maintenance flush after left idle for 30 seconds. The system reset its internal timer afterward. No overflow or pump overheating was observed, confirming the effectiveness of the time-based feature.

Advantages

- Prevents long-term stagnation and odor formation.
- Reduces corrosion risk from hydrogen-sulfide gas.
- Maintains readiness of pumps and sensors.
- Adds reliability in real residential applications.

Summary

The software design successfully implements real-time sensing, automated control, backup activation, and user feedback through simple yet effective embedded programming. The modular code structure allows for future scalability, such as incorporating GSM alerts or data logging for advanced monitoring applications.

3.7 Personal Contribution

TEAM MEMBER	TASK RESPONSIBILITY
Fuad-Nasirudeen Abdussalam	Circuit design, Simulation and Programming in Arduino IDE
Olalekan Abdulfatai Lanre	Hardware Setup and System assembly
Salihu Abdulkadir	Component sourcing and Data testing
Yahya Abdulmuiz Ishola	Troubleshooting, Documentation and report writing
Yakubu Abdurrahman Baba	Prototype construction and System assembly

RESULTS

4.1 Introduction

This chapter presents and discusses the results obtained from the design, implementation, and testing of the Auto Evacuation System for Basement Waste. The results demonstrate how the system responds to different waste levels, how efficiently it transfers waste between containers, and how effectively it detects and handles pump failures.

The testing phase was conducted on a scaled-down laboratory prototype consisting of two transparent plastic containers representing the septic tank and the soak away. The ultrasonic sensor, mounted above the septic tank container, continuously measured the liquid level and sent data to the Arduino Uno microcontroller. The output from the Arduino was displayed on the 16×2 LCD module, showing the waste level in percentage and the operational status of both pumps.

During the tests, various scenarios were simulated to evaluate the performance and reliability of the system, including:

1. Automatic Activation Test — observing how the system reacts when the liquid reaches the upper threshold (70%).

2. Deactivation Test — confirming whether the pumps stop when the level drops below the lower limit (20%).

3. Backup Pump Activation Test — testing how the system detects and responds to a failure of the primary pump.

4. Manual Override Test — verifying that the user can manually operate the pumps through the push button.

5. Error Display Test — checking the LCD for correct indication of system faults or abnormal levels.

6. Timer Activation Test – confirming whether the pumps activate after a predetermined time if the minimum threshold of the septic tank has been surpassed.

Each test was carried out multiple times to ensure consistency and to fine-tune the code and hardware response. The results were recorded, analyzed, and compared with expected outcomes.

The results obtained demonstrate that the system successfully performs automatic evacuation, handles pump faults intelligently, and displays relevant information to the user in real time. The next section (4.2) presents the detailed test observations, data analysis, and corresponding performance discussion.

4.2 Observation and Results

4.2.1 Level Detection Test

The ultrasonic sensor (HC-SR04) was calibrated by measuring the distance from the sensor to the liquid surface and comparing it with a manual ruler measurement.

Table 1

Trial No.	Actual Distance (cm)	Sensor Reading (cm)	Error (%)
1	10	10.4	4.0
2	15	15.3	2.0
3	20	19.7	1.5
4	25	25.5	2.0
5	30	30.2	0.7

Observation:
The sensor maintained an average accuracy of 2% error, which is acceptable for this setting.
The readings were stable when the liquid surface was calm but fluctuated slightly during turbulence caused by pump operation or outside disturbance.

4.2.2 Automatic Pump Activation Test

The main pump was tested by filling the septic tank gradually until the ultrasonic sensor reading reached the high-level threshold (10 cm).
The Arduino activated the relay module, which powered the pump to transfer liquid to the soak-away container.

Table 2

Parameter	Measured Value
Activation level (threshold)	10 cm from sensor
Pump start delay	0.5 s
Pump run time (full evacuation)	~12 s
Stop level (low-level threshold)	4 cm from sensor
Power consumption (12 V pump)	1.2 A average

Observation: The pump responded almost instantly when the liquid reached the set threshold and successfully lowered the level to the stop point without human intervention.

4.2.3 Backup Pump Activation Test

To simulate a **main pump failure**, the primary pump wire was temporarily disconnected.
When the level reached the high threshold, the Arduino attempted to activate the main pump but detected no level drop within the expected timeout (10 s).
The system then switched automatically to the backup pump.

Table 3

Test Condition	Result
Primary pump disconnected	Backup pump activated after 10 s
Level drop after backup activation	Successful (12 s)
Fault indicator LED	ON
System recovery	Automatic (LED reset after next cycle)

Observation: The system accurately detected pump failure and engaged the backup pump as intended.

4.2.4 Time-Based Flush Test

To evaluate the **time-based flushing module**, the tank was left at half capacity without triggering the high-level sensor. The Arduino was programmed to initiate a maintenance flush after **1 minute** (for demonstration purposes, scaled down from the actual 14 days).

Table 4

Condition	Expected Behavior	Observed Result
No pump operation for 1 min	System should start short flush	Activated after 60 s
Pump run duration	15 s	15 s
Backup pump use (if fault detected)	Switched correctly	Passed
LED indicator	Blinked during maintenance flush	Correct

Observation: The time-based module performed as expected, automatically activating a short flush cycle after the preset period. This feature effectively prevents waste stagnation during long idle periods, validating the design's additional reliability feature.

4.3 Discussion of Results

The developed Auto Evacuation System for Basement Waste was successfully built and tested to verify its performance, reliability, and response under different operating conditions. The system was implemented as a small-scale prototype using two transparent containers to represent the septic tank and soak away. The ultrasonic sensor was mounted on top of the septic tank container, while two DC pumps were installed near its base for automatic waste transfer. The entire system was controlled by an Arduino Uno microcontroller, which processed sensor data and executed the control logic.

During testing, the operation of the system was observed under various simulated conditions to assess how effectively it performed its automated functions.

System Operation Test

When the water level in the septic tank container was increased, the ultrasonic sensor accurately measured the distance and converted it into a level percentage. Once the level reached approximately 70%, the microcontroller activated Pump 1, which began transferring the liquid to the soak away container. The level gradually reduced as the pump operated. When the liquid level fell below 20%, the system automatically stopped the pump to prevent dry running.

This test confirmed that the level detection and pump control logic were functioning as designed. The readings displayed on the LCD were stable and corresponded well to the actual fluid level, indicating the sensor's precision and calibration accuracy.

Backup Pump and Fault Detection

To evaluate the system's fault detection and redundancy feature, the first pump's connection was deliberately interrupted to simulate failure. The system automatically detected that the waste level was not decreasing after activation of Pump 1. Within a few seconds, the Arduino controller switched OFF the first pump and started the second pump (Pump 2). Simultaneously, an error message appeared on the LCD to notify the user of Pump 1's malfunction.

This test validated the reliability and intelligence of the control algorithm. The backup mechanism ensured that evacuation continued smoothly, minimizing the chances of overflow or operational downtime.

Manual Override Test

The manual control button was pressed to verify the system's manual override functionality. Regardless of the sensor readings, both pumps were activated immediately when the button was pressed. This function is especially useful for maintenance purposes or in situations where the automatic mode needs to be temporarily bypassed. The system returned to automatic operation once the button was released, demonstrating seamless transition between manual and automatic modes.

Time-Based Activation

In addition to level-based control, the software also included a time-based evacuation function. The Arduino was programmed to activate the pump at fixed intervals even when the waste level remained constant. This feature was tested successfully — the pump automatically started after a set period (three minutes), ran for a short duration, and then turned off. This function prevents stagnation and ensures that no chemical buildup occurs in the septic tank when usage is low.

General Performance Discussion

Overall, the system performed as expected across all test conditions. The ultrasonic sensor provided accurate and stable readings, while the Arduino controller executed the programmed logic efficiently. The system responded quickly to level changes, activated the correct pump when necessary, and displayed clear status information on the LCD. The backup pump feature proved vital in maintaining continuous operation during simulated faults, and the manual override button provided convenient operator control.

The successful integration of hardware and software demonstrates the effectiveness of using microcontroller-based feedback control for waste management systems. The project achieved its objectives by automating basement waste evacuation, reducing human intervention, and ensuring environmental safety.

CONCLUSION

5.1 Conclusion

The design and implementation of the Auto Evacuation System for Basement Waste successfully achieved the intended objectives of automating waste evacuation from an underground septic tank. The project effectively demonstrated how sensing, control, and actuation can be combined using affordable and accessible electronic components to address a real-world engineering problem.

The developed system uses an ultrasonic sensor to monitor the waste level in real time and an Arduino Uno microcontroller to process the data and control two pumps through motor driver circuits. The control algorithm automatically activates the primary pump when the waste level reaches a specified upper threshold and stops it when the level falls below the lower threshold. In case of a pump malfunction, the system intelligently switches to a secondary pump, ensuring continuous operation and preventing overflow.

The LCD display provides real-time feedback to the user, including the current waste level, pump status, and error messages. The manual override feature allows the user to operate the pumps directly, when necessary, while the time-based trigger ensures periodic evacuation even in low-usage conditions.

Testing results confirmed that the system performed reliably under all simulated scenarios. It responded quickly to level changes, executed control actions accurately, and maintained stable operation throughout the test period. The inclusion of fault detection and redundancy enhanced the system's safety and dependability, making it suitable for real-world applications in residential and commercial buildings with basements.

In conclusion, the project demonstrates a practical and cost-effective solution for automating basement waste evacuation. It promotes improved hygiene, reduces manual labor, prevents septic tank damage, and contributes to environmental safety. The successful outcome of this project highlights the potential of microcontroller-based systems in solving engineering challenges through intelligent automation.

5.2 Further Work

Although the prototype of the Auto Evacuation System for Basement Waste successfully met its primary design objectives, there are several areas where the system can be improved and expanded to enhance performance, reliability, and practicality for real-world applications.

1. <u>Integration of GSM or IoT Communication:</u>

The system can be upgraded to include a GSM module or Wi-Fi connectivity, enabling it to send SMS alerts or transmit data to a mobile device or web dashboard. This would allow remote monitoring of the septic tank level and system status without requiring physical presence at the site.

2. Automatic Data Logging and Analysis:

Incorporating a data logging feature using an SD card or cloud-based platform would enable storage of historical data such as pump activation times, waste levels, and system errors. This information could help in analyzing long-term trends and optimizing maintenance schedules.

3. Improved Power System and Energy Efficiency:

The use of solar panels or battery backup systems could make the project more sustainable and suitable for off-grid locations. Additionally, incorporating power-saving modes in the software could reduce energy consumption when the system is idle.

4. Use of Industrial-Grade Components:

For large-scale or permanent installations, the pumps, sensors, and wiring can be replaced with industrial-grade materials that withstand harsh conditions, including corrosion, humidity, and high waste viscosity.

5. Enhanced User Interface:

A graphical interface could be implemented using a touchscreen LCD or a mobile application to provide users with a more interactive control and monitoring experience. This interface could include graphical indicators, historical trends, and real-time error notifications.

6. Integration of Safety Alarms and Backup Storage:

The system could include audible or visual alarms that alert occupants when the waste level exceeds a dangerous threshold or if both pumps fail simultaneously. A temporary holding chamber could also be incorporated to handle overflow conditions.

7. Advanced Control Algorithms:

Implementing PID (Proportional-Integral-Derivative) or fuzzy logic control algorithms could make the evacuation process more adaptive to changing waste inflow rates, improving response accuracy and efficiency.

Summary

By incorporating these enhancements, the Auto Evacuation System for Basement Waste can evolve from a prototype demonstration into a fully functional, intelligent, and sustainable solution for modern waste management systems. The proposed improvements would make it more reliable, energy-efficient, and user-friendly, thereby increasing its suitability for deployment in residential, commercial, and industrial applications.

REFERENCES

[1] U.S. Environmental Protection Agency (EPA), Onsite Wastewater Treatment Systems Manual, EPA/625/R-00/008, 2002.

[2] Oregon Department of Environmental Quality (DEQ), STEP/STEG Design Guidelines, 2018. Available online: https://www.oregon.gov/deq/

[3] Penn State Extension, Septic Tank Pumping, 2021.
Available online: https://extension.psu.edu/septic-tank-pumping

[4] DiStefano, J. J., Stubberud, A. J., and Williams, I. J., Schaum's Outline of Feedback and Control Systems, McGraw-Hill, 1997.

[5] MaxBotix Inc., Advantages and Limitations of Ultrasonic Sensors, Technical Note, 2019. Available online: https://www.maxbotix.com/

[6] FluidSwitch Technologies, The Disadvantages of Ultrasonic Liquid Level Sensors, 2020. Available online: https://fluidswitch.com

[7] Susilawati, S., et al., The Design of Arduino Prototype for Monitoring Septic Tank Wastewater Levels, International Journal of Advanced Research in Engineering and Technology (IJARET), vol. 11, no. 8, pp. 356–365, 2020.

[8] Romtec Utilities, The Importance of Redundancy in Pump Station Engineering, 2021. Available online: https://romtecutilities.com/

[9] North Carolina Department of Health and Human Services (NCDHHS), Design Criteria for Septic Tank Effluent Pumps, 2017.
Available online: https://epi.dph.ncdhhs.gov/

[10] Bayero University Kano, Department of Mechatronics Engineering, MCS4209

Project Report Template, Faculty of Engineering, 2022.

[11] Arduino. (2024). *Arduino UNO R3 Technical Specifications. Available online at: https://www.arduino.cc

[12] HC-SR04 Ultrasonic Sensor Datasheet. (2023). Elecrow Electronics

Documentation. Available online at:

https://www.elecrow.com

[13] DiStefano, J. J., Stubberud, A. J., & Williams, I. J. (1997). Schaum's Outline of Feedback and Control Systems. McGraw-Hill Professional Publishing.

[14] Petruzella, F. D. (2010). Programmable Logic Controllers (4th ed.). McGraw-Hill Education.

[15] Kumar, R., & Mehta, S. (2020). “Microcontroller-Based Sewage Level Monitoring and Alarm System.” International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering, Vol. 9, Issue 5, pp. 1123–1130.

[16] Joseph, T., & Abraham, J. (2018). “Design and Implementation of Automatic Water Level Controller Using Arduino.” International Journal of Engineering Research & Technology (IJERT), Vol. 7, Issue 8, pp. 1–4.

[17] Patel, A., & Singh, D. (2019). “Low-Cost Water Tank Level Monitoring System Using Ultrasonic Sensor and Arduino.” International Research Journal of Engineering and Technology (IRJET), Vol. 6, Issue 3, pp. 2012–2016.

[18] Rahman, M., Das, S., & Chowdhury, K. (2021). “Smart Drainage Cleaning and Water Level Control System.” International Journal of Innovative Science and Research Technology, Vol. 6, Issue 2, pp. 455–460.

[19] Singh, V., Chatterjee, A., & Pathak, P. (2022). “Dual Pump Control System with Automatic Backup Switching Using Arduino.” International Journal of Engineering Trends and Technology (IJETT), Vol. 70, Issue 4, pp. 90–96.